Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim:** Simulate discrete time sequences.	
Experiment-10	**Date:01-08-2026**	**Enrollment No:92400133079**

Code:-

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

# Read the uploaded image automatically
filename = next(iter(uploaded.keys()))
img = cv2.imdecode(
    np.frombuffer(uploaded[filename], np.uint8),
    cv2.IMREAD_GRAYSCALE
)

if img is None:
    raise ValueError("Could not read the uploaded image. Try another image file.")

print("Image loaded successfully:", filename)
print("Image dimensions:", img.shape)

# Fourier Transform
F = np.fft.fftshift(np.fft.fft2(img))
M, N = img.shape

# Frequency distance matrix
```

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```
u, v = np.meshgrid(
    np.arange(N) - N // 2,
    np.arange(M) - M // 2
)
D = np.sqrt(u**2 + v**2)

# Low-pass and high-pass filters
LPF = D < 30
HPF = D >= 30

# Apply filters
smooth = np.abs(np.fft.ifft2(np.fft.ifftshift(F * LPF)))
high_pass = np.abs(np.fft.ifft2(np.fft.ifftshift(F * HPF)))

# Sharpen image
sharp = np.clip(img.astype(float) + high_pass, 0, 255).astype(np.uint8)
smooth = np.clip(smooth, 0, 255).astype(np.uint8)

# Display results
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(smooth, cmap='gray')
plt.title('LPF - Smoothing')
```

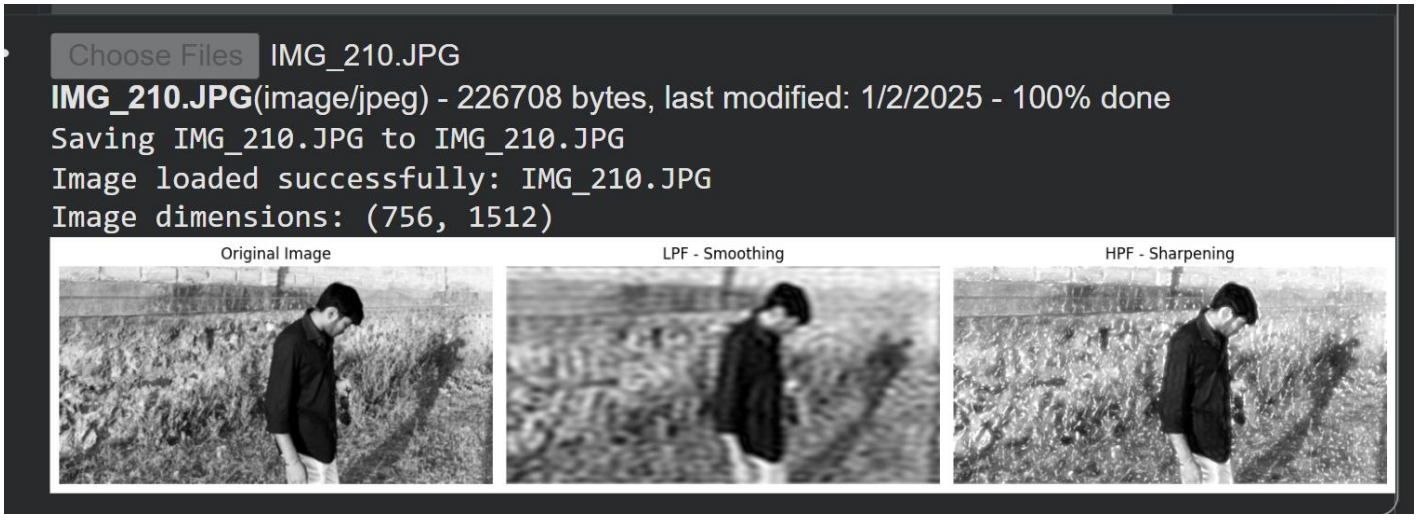
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```
plt.axis('off')


plt.subplot(1, 3, 3)
plt.imshow(sharp, cmap='gray')
plt.title('HPF - Sharpening')
plt.axis('off')


plt.tight_layout()
plt.show()
```

output:-



Github : https://github.com/yugmakwana99/DSIP.git